

Optimizing agrivoltaic systems: A comprehensive analysis of design, crop productivity and energy performance in open-field configurations

Abstract

The rapid expansion of photovoltaic (PV) systems is essential for decarbonizing energy systems but raises concerns about competing with agriculture for arable land. Agrivoltaic (APV) systems offer a sustainable solution by integrating PV installations with agriculture on the same land. However, adoption is constrained by a limited understanding of design parameters and their effects on energy and crop productivity. This study presents a simulation framework to evaluate open-field APV systems, focusing on the impacts of ground coverage ratio (GCR), clearance height, and tracking configurations on solar irradiance distribution, crop yield, and solar energy generation. The Land Equivalent Ratio (LER) was employed to assess the combined productivity of agricultural output and solar energy generation. Scenarios are modeled for temperate climates, examining three crops (lettuce, turnip, and maize) alongside bifacial PV systems with fixed and tracking designs. Results showed that design parameters significantly affect shading and light distribution. Higher clearance heights and lower GCRs enhance agricultural productivity, particularly for shade-sensitive crops like maize, though slightly reducing energy yields. LER values ranged from 1.04 to 2.05 for the GCR variable between 0.6 and 0.2, demonstrating that APV systems are more productive than conventional land use. These insights support optimizing APV designs, aiding policymakers and developers in aligning renewable energy goals with agricultural sustainability.

1. Introduction

The increasing global demand for energy and the pressing need to decarbonize energy systems have positioned renewable energy technologies at the center of climate change mitigation strategies. Among these technologies, photovoltaic (PV) systems have gained prominence due to their scalability, efficiency, declining costs, and modular nature of their deployment. According to the International Energy Agency (IEA), solar PV installations are set to triple by 2030, accounting for a significant portion of global renewable electricity generation. However, the increasing deployment of PV systems raises concerns regarding land use, as utility-scale ground-mounted PV installations often compete with agricultural land, creating a potential trade-off between energy and food production [1]. Several strategies have been proposed to mitigate this issue. One approach is the integration of PV systems on the roofs of existing buildings, reducing the need for additional land occupation. However, rooftop PV installations are constrained by surface availability and structural limitations, making large-scale implementation challenging [2].

Agrivoltaics (APV), developed in 1981, presents a complementary and scalable solution by co-locating energy generation and agricultural activities on the same land [3]. Unlike conventional ground-mounted PV systems, APV configurations minimize the impact on food production while optimizing solar energy capture. Over the past few decades, APV systems have seen rapid adoption in countries such as Japan, France, Germany, and China, with a variety of applications ranging from vineyards to arable crops [4], [5]. Recent studies have demonstrated that APV installations can enhance land-use efficiency, particularly in regions with high solar irradiance and agricultural productivity potential. By adjusting design parameters such as ground coverage ratio (GCR), clearance height, and tracking configurations, APV systems can maintain crop yield while generating significant renewable energy [6]. This dual-use approach aligns with sustainable development goals, promoting land conservation while advancing the energy transition. The importance of APV systems in accomplishing sustainable development goals is further highlighted by the introduction of regulatory incentives, such as those found in the Common Agricultural Policy (CAP) of the European Union [7]. Understanding and reducing the trade-offs between crop productivity and energy generation have been the main focus of academic research. Studies like those by Mamun et al. [5] and Trommsdorff et al. [8] have examined the effects of shade, the distribution of solar irradiance, and the resulting crop-specific yield variations. Katsikogiannis et al.

optimized APV designs for diverse climates and crops [9]. Their study analyzed three bifacial APV topologies in Boston: S-N facing, E-W wings, and E-W vertical, improving specific yield by 39 %, 18 %, and 13 %, respectively, compared to monofacial PV. The E-W vertical suits permanent crops, while the S-N facing requires shade-tolerant species in summer. The E-W wings design balances shading and light distribution, increasing land productivity by 50 % but reducing electrical yield by 33 %. Their findings support APV adoption for sustainable agriculture. The Land Equivalent Ratio (LER) is a key indicator in APV systems, as it measures land productivity [6]. When LER is greater than 1, it indicates that dual land use (PV and agriculture) is more efficient. When LER is less than 1, it suggests that separate land use for PV installation and agriculture might be more effective. Reher et al. found that APV systems in Belgium reduced crop yields (11 %–46 %) but achieved slightly positive LER, suggesting potential for renewable energy generation along [10]. Technological developments in PV system designs, like tracking systems and bifacial panels, have opened up new possibilities for improving land-use efficiency and increasing light availability. Berrian et al. developed an integrated tool to optimize APV tracking systems by modeling both crop and PV performance [6]. Evaluating a bifacial APV system in Germany, they found that row spacing significantly affects land productivity, while increasing tracker height reduces the feasibility of dual land use. Adjusting tracker orientation improves productivity, but a 40 % PV module density leads to a 42 % energy loss and substantial crop yield reductions (32 % for potatoes, 66 % for maize). They also introduced simplified crop yield models for seamless integration into PV design software. An important consideration for the long-term performance of APV systems is the influence of environmental factors such as dust accumulation, wind storms, and heavy rainfall. Dust deposition and bird droppings on PV modules can lead to a notable decrease in energy output, with studies indicating reductions ranging from 2 % to 50 %, depending on regional climatic conditions and cleaning frequency [11], [12]. In arid and semi-arid regions, where APV adoption is growing, regular panel maintenance strategies are essential to mitigate soiling losses [13].

Furthermore, wind storms and extreme weather events may lead to structural stresses on both PV arrays and crops, in a context in which climate change could have direct effects on PV generation [14]. Heavy rainfall, while beneficial for the natural cleaning of panels, can cause localized flooding, potentially affecting crop yield and soil stability [15].

Despite these advancements, research on APV systems remains fragmented. Most studies are either geographically limited, focusing on specific climatic conditions, or crop-specific, addressing particular agricultural scenarios. Additionally, the lack of standardized design practices, coupled with the variability in environmental and regulatory contexts, presents challenges for wider implementation.

Although the benefits of APV systems are well-documented, significant knowledge gaps remain. First, the interplay between design parameters—such as the GCR, structure height, and the use of tracking mechanisms—and their combined impact on energy and agricultural yields is poorly understood. Existing studies often focus on isolated parameters or fail to capture the full complexity of integrated APV systems [8], [16].

Second, there is limited research on how APV systems perform across diverse crop types, particularly in temperate climates where seasonal variations significantly influence both solar energy generation and crop growth. While certain crops, like shade-tolerant leafy vegetables, have been studied extensively, the performance of shade-sensitive crops, such as cereals or maize, remains underexplored [17], [18].

Lastly, the lack of a comprehensive modeling framework that integrates solar irradiance distribution, crop yield simulations, and energy performance metrics limits the ability of researchers and practitioners to design optimized APV systems. This gap is particularly evident in studies addressing open-field configurations, where variability in design and environmental factors creates a pressing need for robust simulation tools and sensitivity analyses [8], [19].

This study aims to bridge these gaps by developing a comprehensive framework for designing and evaluating open-field APV systems. The specific objectives include:

- Integrated sensitivity analysis for temperate climates: while previous studies have examined shade sensitivity and LER in APV systems, this research work provides a comprehensive sensitivity analysis of key design parameters (GCR, clearance height, and tracking) specifically for temperate climates. Many prior studies have focused on high solar irradiance regions, while our results provide valuable insights for regions with seasonal variations.

- Enhanced modeling framework with crop-specific responses: the modeling framework employed in this study is widely used, but the integration of detailed crop yield response curves for multiple crops under APV conditions proposed in this study refines existing models. In particular, a crop-specific assessment of land-use efficiency is provided, which is often generalized in prior works.

- Optimization of APV design for specific crop categories: this research study introduces a multi-crop comparative framework, analyzing shade-benefiting, shade-tolerant, and shade-sensitive crops in a single analysis. Unlike previous works that focus on individual

crops, the proposed approach provides generalizable design recommendations based on crop categories rather than specific species.

Policy-relevant insights for APV adoption in Europe: findings can be translated into actionable guidelines for APV system design in European agricultural contexts. This is particularly relevant given the increasing regulatory incentives in the EU for APV adoption.

By addressing these objectives, this research aims to contribute to the development of standardized design practices and decision-support tools for APV systems, supporting their broader adoption as a sustainable land-use solution.

2. Methodology

This section describes the simulation framework developed to assess the performance of APV systems, including the mathematical and physical foundations of the individual submodels used to calculate the performance measures. The study of APV systems, from the assessment of solar radiation to the comparison of productivity outputs, requires the implementation of the submodels described herein and is needed for the assessment of irradiance, crop yield, energy yield and APV performance metrics.

2.1. Bifacial radiance and irradiance assessment

Irradiance distribution was simulated using Bifacial Radiance (BR), a Python-based tool for modeling PV systems. This software integrates the Radiance ray-tracing engine to calculate the spatial distribution of solar radiation on both PV modules and the ground. The BR framework follows an object-oriented structure, where different classes define key components of the simulation, ensuring modularity and flexibility in analyzing various system configurations.

The primary classes in BR include:

1.

RadianceObj: This class serves as the central controller for Radiance simulations, managing key elements such as input files, sky conditions and simulation parameters. It initializes the Radiance environment, sets up scene lighting conditions based on meteorological inputs, and controls the execution of ray-tracing calculations. Additionally, it allows the user to define different sky models, including clear-sky and

overcast conditions, optimizing the irradiance distribution analysis for diverse environmental scenarios.

2.

GroundObj: This class is responsible for defining the optical properties of the ground surface, including its reflectivity (albedo) and material characteristics. Since bifacial PV modules capture irradiance from both direct sunlight and reflected light from the ground, accurately modeling the ground's optical behavior is crucial. GroundObj enables customization of surface reflectance based on different soil types, vegetation cover, or artificial ground treatments, improving the precision of bifacial gain estimations.

3.

SceneObj: This class defines the spatial configuration and geometric layout of the PV array, including parameters such as module tilt, row spacing, clearance height, and tracking mechanisms. SceneObj plays a fundamental role in determining the level of mutual shading between modules and the extent of irradiance reaching the ground.

4.

MetObj: This class processes meteorological data, including global horizontal irradiance (GHI), direct normal irradiance (DNI), diffuse horizontal irradiance (DHI), air temperature, and wind speed. It extracts data from standard meteorological databases such as EnergyPlus Weather (EPW) files and Typical Meteorological Year (TMY) datasets. By dynamically integrating weather conditions into the simulation, MetObj allows for seasonal and hourly analysis of system performance, capturing variations in irradiance due to atmospheric conditions.

5.

AnalysisObj: This class handles the post-processing of simulation results, including the generation of irradiance heatmaps, statistical reports, and comparative visualizations. It enables users to analyze irradiance distribution at different spatial points within the system, quantify shading losses and compute bifacial gain. Additionally, AnalysisObj includes built-in functions for exporting data in formats compatible with external modeling tools, facilitating further performance evaluations and integration with crop growth models.

These modular classes collectively form a robust simulation environment for analyzing APV system performance, allowing for a flexible and detailed assessment of solar irradiance distribution under various design and environmental conditions.

Simulations involved 12 Typical Meteorological Days (TMDs) using a new *groundscan* function to place 300 virtual sensors over a grid area of about m, avoiding border effects. The simulated area was centered and covered at least ten PV modules along rows

and twice the collector width in spacing. Detailed scene geometry is introduced in [Section 3](#).

Radiance calculations utilized *gendaylit* and *rtrace*. *Gendaylit*, based on Perez models [\[20\]](#), [\[21\]](#), calculates solar angles and radiation using DNI, DHI, and albedo inputs. It generates angular distributions of daylight sources using the equations: (1) where θ_z is zenith luminance, a_1, a_2, a_3, a_4 and a_5 are coefficients dependent on solar altitude, clearness, and brightness [\[20\]](#), [\[21\]](#). These models handle all-weather sky types, offering flexibility beyond the default *gensky* tool. The luminous efficacy model [\[20\]](#) converts the radiation to illuminance with an efficacy factor.

Rtrace implements a hybrid deterministic/stochastic ray tracing technique, efficiently modeling irradiance interactions using an octree spatial subdivision scheme [\[22\]](#). This approach calculates irradiance for bifacial PV modules and structural shading with Kajiya's rendering equation [\[23\]](#): (2) where R_r, R_e, R_i are reflected, emitted, and incident radiances, and f_r is the bidirectional reflectance distribution function (BRDF) [\[24\]](#). A hybrid solution optimizes bias-variance trade-offs, enhancing efficiency [\[25\]](#), [\[26\]](#). While view factor techniques are viable alternatives, ray tracing better addresses bifacial irradiance complexities.

2.2. Reduction in solar radiation and crop yield assessment

Crop yield is primarily influenced by light availability and land occupation constraints imposed by PV structures. Instead of using complex mechanistic crop growth models, a meta-regression approach was used to estimate crop yield based on Relative Plot Yield (RPY) as a function of the Reduction in Solar Radiation (RSR), following [\[27\]](#), [\[28\]](#), [\[29\]](#). RSR, derived from BR ground scans ([Section 2.1](#)), represents the ratio of irradiance received by crops under APV versus open-field conditions.

Crop data, pooled from intercropping and artificial shading studies, cover nine crop types: berries, fruits, vegetables, leafy vegetables, C3 cereals, maize, tubers/root crops, grain legumes, and forages. Results reveal a nonlinear relationship between yield and RSR, with most crops tolerating up to 15 % RSR without significant losses. However, maize and grain legumes show substantial yield reductions under even modest shading [\[27\]](#).

Crop response to shading is classified into three categories [\[30\]](#):

- **Shade-benefiting:** Yield improves under low shading but declines at higher RSR.

-
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Shade-tolerant: Yield decreases less than proportionally with shading.

Shade-sensitive: Yield drops significantly under shading.

For example, forages are shade-benefiting up to 25 % RSR and shade-tolerant at higher RSR, while C3 cereals are shade-tolerant until 50 % RSR but shade-sensitive beyond that. The RPY for each crop is then adjusted for land occupation constraints using the lost land ratio:(3)The revised yield, Relative Field Yield (RFY), incorporates both RPY and :(4)

Additional factors like light homogeneity and microclimate changes beneath PV modules further influence yields. Shadows cast by PV modules vary daily and seasonally, affecting solar radiation distribution critical for crop growth, uniform ripening, and synergy effects. Moreover, microclimate alterations, such as temperature and humidity variations, may benefit or hinder specific crops.

2.3. PVsyst and energy yield assessment

The energy yield was computed using PVsyst, a versatile tool for simulating solar energy systems. PVsyst enables detailed modeling of system components, including PV modules and inverters, to accurately estimate performance under varied conditions. PVsyst's models are categorized into meteorology and PV system components. Meteorological models calculate solar geometry and irradiance on the collector plane using time-averaged variables. The transposition of irradiance from horizontal to tilted planes employs the Hay and Perez models [20]. This study uses the Perez model, which incorporates isotropic, circumsolar, and horizon band components, achieving greater reliability through accurate horizontal irradiance data. Effective incident irradiance is calculated by factoring in shading losses and the Incidence Angle Modifier (IAM). For bifacial PV modules, rear-side irradiance is modeled using view factor techniques. As far as the PV system components, the behavior of the system can be simulated by inserting the specifications of the main components in the software for calculation, with particular regard to PV module specifications. Shockley's 5-parameter single-diode model [31] is used to compute the I–V curve for given irradiance and temperature conditions. This model was developed for a PV cell but can be generalized to a PV module if all cells are considered identical. Once the calculation is extended to the electrical connection of PV modules in series and/or parallel, the calculation is extended to other main components such as inverters and transformers to eventually obtain the power output of the system.

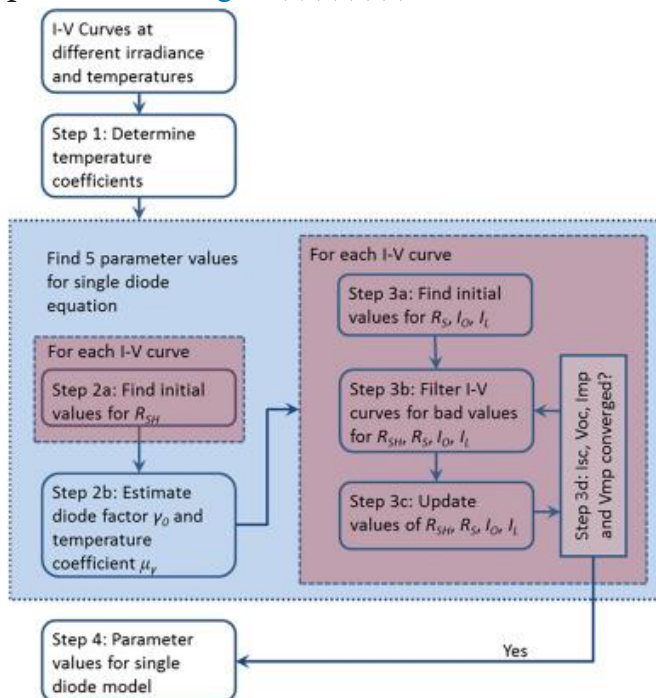
The equation that describes the single-diode model for a given set of reference conditions is expressed as:(5)

where I_{ph} is the photo-generated current (A), I_0 is the dark saturation current (A), n is the diode ideality factor (unitless), V_t is termed the thermal voltage (V) for the module, k is Boltzmann's constant (J/K), q is the elementary charge (coulomb), R_s is the series resistance (Ω), R_{sh} is the shunt resistance (Ω).

The quantities I_{ph} , I_0 , n , and V_t are frequently termed the "five parameters" or "five variables" and they depend on the exogenous quantities G and T .

The following auxiliary equations are introduced as well:(6)(7)(8)(9)(10)(11)

In these equations, the subscript ref indicates a value at the reference conditions and G_{eff} for effective irradiance and cell temperature, respectively; typical values are 1000 W/m^2 and 300 K . Other parameters appearing are α is the temperature coefficient for I_{ph} (A/K), E_g is the effective band gap (eV), β is the temperature coefficient for the diode ideality factor (1/K), R_{sh0} is the shunt resistance in the absence of irradiance (Ω), R_{shref} is the shunt resistance at reference irradiance (Ω), γ is an empirical term describing the change of shunt resistance with irradiance (unitless). The following parameters are estimated according to the scheme presented in Fig. 1: I_{ph} , I_0 , n , V_t , R_s , R_{sh} , and γ .



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Fig. 1. Scheme for single-diode model parameter estimation process.

Parameter estimation proceeds beginning with the measurement of a set of I–V curves over a range of irradiance and cell temperature conditions. Next, values are determined for each I–V curve for the five parameters in the single-diode equation. Finally, parameter values are determined for the auxiliary equations that complete the single-diode model.

This study employs the JAM78D40-600 module by JA Solar, featuring 600 bifacial double glass technology and n-type half-cell configuration. Component data and default settings are used to estimate energy yields for the analyzed configurations.

2.4. Land equivalent ratio and performance metrics

The Land Equivalent Ratio (LER) metric is introduced to assess the performance of the APV system and compare the case studies. It is a key indicator first used by Dupraz et al. [32] and gradually applied in most APV study cases, and is used to compare the conventional approach (separate PV and crop systems) with the integrated solution on the same land area. LER determines whether the combined value of agricultural production and solar energy generation is equivalent to or greater than the value of land when used alone. It is computed as: (12) where the first ratio corresponds to , indicating crop relative yield and also equal to RFY in the context of this study, while the second ratio corresponds to indicating PV relative yield.

3. Case study

Integrating PV systems with agriculture presents challenges requiring design adaptations. To address these, the described models are applied to simulate representative open-field APV systems, combining ground-mounted PV systems with crop production.

3.1. Location

The case study focuses on Tuscania (VT), Italy (42.37N 11.70E), an area of interest for utility-scale developers due to favorable land and climate conditions, see Fig. 2. According to the Italian Energy Service Operator (GSE), Viterbo province indeed emerges as one of the provinces with the highest installed PV capacity and highest capacity per installed plant.



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Fig. 2. Satellite view of Tuscany (VT).

In this study, meteorological data were sourced from the PVGIS SARA2 database [33], [34], a widely used platform providing high-resolution, satellite-derived solar radiation data for Europe, Africa, and parts of Asia. PVGIS is particularly suitable for PV performance assessments due to its integration of ground-based measurements from the European Solar Radiation Atlas (ESRA) and its validated methodologies. However, it is essential to acknowledge potential uncertainties associated with different meteorological datasets.

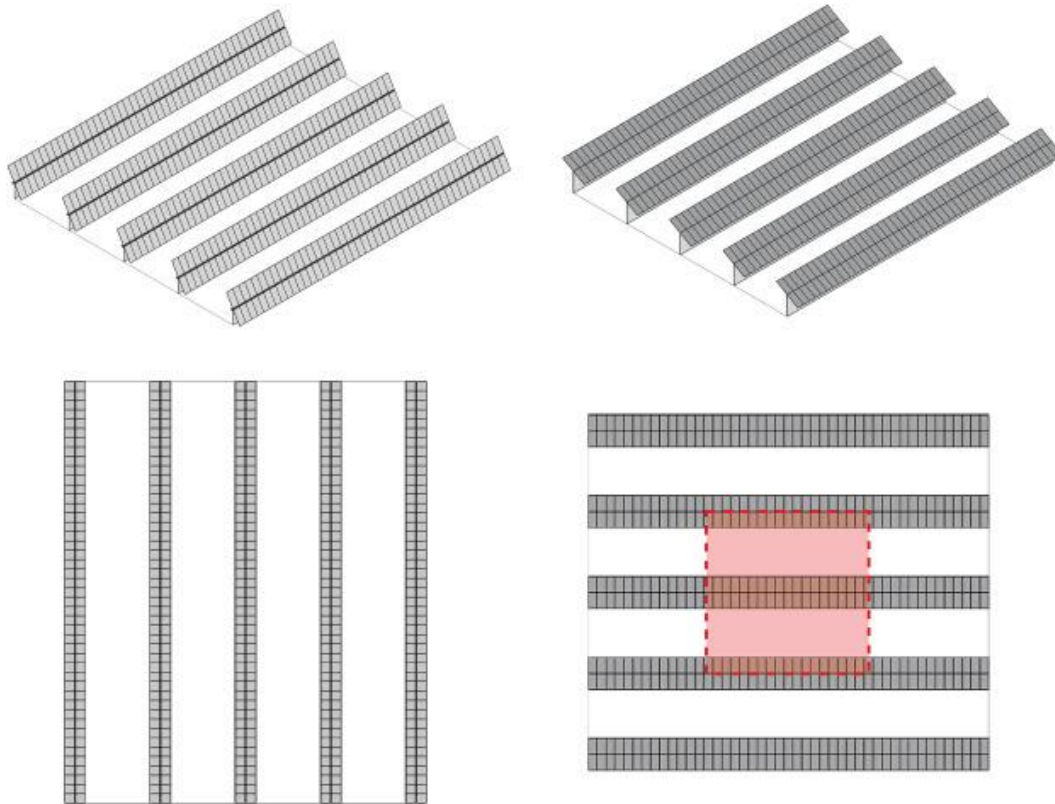
Alternative data sources, such as Renewables.ninja [35] and RETScreen [36], offer distinct methodologies for solar radiation estimation. Renewables.ninja employs a combination of satellite imagery and reanalysis data (e.g., MERRA-2) to model solar irradiance, potentially reducing bias in regions with complex atmospheric conditions. RETScreen, on the other hand, utilizes NASA's meteorological dataset, incorporating ground-based observations with a focus on long-term climatology. Comparisons have shown that PVGIS tends to slightly underestimate summer energy production while being more accurate in annual predictions, whereas RETScreen exhibits higher deviations due to its reliance on global-scale models with coarser resolution [37].

The choice of PVGIS was made based on its validation against real-world PV installations and its accessibility for high-resolution, site-specific solar radiation estimates. Nevertheless, future work should incorporate a sensitivity analysis using multiple data sources to quantify the impact of dataset selection on APV system performance assessments. Meteorological data, obtained from PVGIS SARA2, uses the TMY format, comprising hourly values for variables like date and time, global horizontal irradiance (GHI), direct normal irradiance (DNI), diffuse horizontal irradiance (DHI), air pressure, dry bulb temperature, wind speed, wind direction, relative humidity, longwave downwelling infrared radiation. The most relevant data are extracted and reported in the supplementary file. The selected format, TMY (Typical Meteorological Year), is

produced by the database provider and includes hourly data from the most typical months out of the multi-year time series available, e.g., 16 years (2005–20) in the case of PVGIS-SARAH2 database. To streamline computations, the dataset is reduced to TMDs, representing 12 typical days per year, as illustrated in the supplementary file, following the method of [38].

3.2. PV system

Two PV configurations are analyzed: Tracker 2P and Fixed 2P. Tracker 2P includes two modules in portrait orientation, with an azimuth angle of 90 and variable tilt, a GCR of 0.5, and a clearance height of 0.5 m. Fixed 2P also features two modules in portrait orientation but with an azimuth angle of 0 and a fixed 25 tilt. It has the same GCR but a higher clearance height of 2.5 m, classifying it as an overhead APV system. These configurations are illustrated in Fig. 3.



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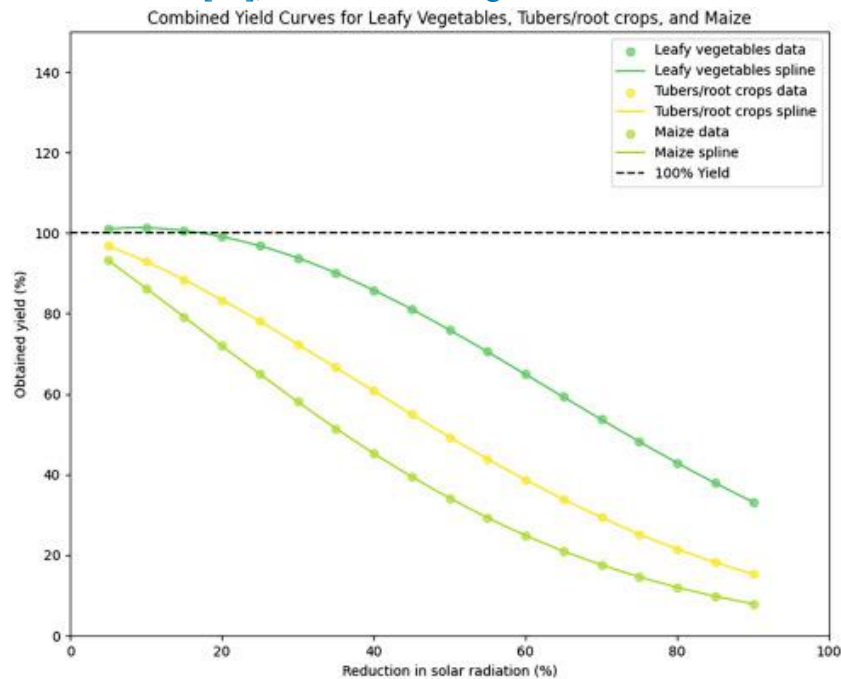
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Fig. 3. Axonometric (top) and top (bottom) views of Tracker 2P (left) and Fixed 2P (right). North is at the top.

For the simulation, 300 sensors are placed over a center grid to assess irradiance distribution. The model accounts for torque tube shading losses, with a 0.2 m gap between modules to minimize shading effects.

3.3. Crop system

Crop systems are evaluated in terms of land use and yield potential. The entire land area is available for overhead systems except for structural footprints, assumed as a 3 % loss. For interspace systems, the lost land increases with GCR and decreases with clearance height. For m and GCR values of 0.2, 0.3, 0.4, 0.5, and 0.6, values are 0.17, 0.26, 0.35, 0.42, and 0.62, respectively. With m , is reduced to 0.03 for all GCR values. The relationship between RSR and crop yield was derived from experimental data, covering various crop types such as leafy vegetables (e.g., lettuce), root crops (e.g., turnip), and cereals (e.g., maize), selected for their diverse shade tolerance behaviors. Yield curves, derived from [39], are shown in Fig. 4.



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Fig. 4. Spline interpolation curve for Lettuce, Turnip, and Maize yield estimation.

Shade-loving crops like lettuce have the lowest Light Saturation Point (LSP) at 213 W/m², benefiting from low light conditions [40]. Turnip exhibits moderate LSP at 469 W/m², balancing shade tolerance and light needs. Maize, a sun-loving crop with an LSP of 685 W/m², thrives under high radiation but is highly sensitive to shading [41].

These crops are suitable for the Tuscania region and represent a range of shading responses, from shade-benefiting to shade-sensitive. Literature confirms shade impacts on growth and yield, with lettuce showing low losses due to adaptive responses [27] and maize being highly susceptible to reduced light availability [42].

4. Results

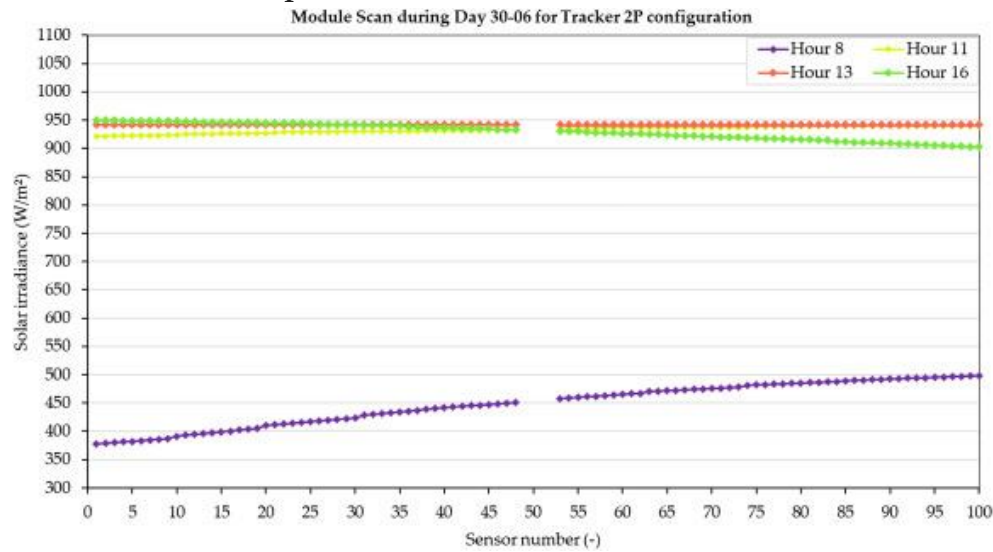
In this section, the most relevant output is selected and extrapolated from the large amount of data obtained as a result of the model simulations on the configurations introduced. In particular, a closer look at the three TMDs for March (17-03), June (30-06), and September (29-09) was taken. These months were selected as spring and summer correspond to the cropping season for many crops and also cover a sufficient range of GHI values.

4.1. Irradiance assessment

A representation of the irradiance levels on PV modules, particularly at ground level, is provided through heat maps utilizing the *viridis* color map. This choice avoids the potential inaccuracies associated with rainbow color maps, which are not only less suitable for precise data interpretation but also inaccessible to color-blind users [43]. These heat maps enable a qualitative assessment of irradiance distribution under typical conditions and across varying time scales. Such a representation is recognized as an effective method for facilitating quick and intuitive visual analysis. The heat maps can employ a fixed unit scale to enable comparisons across different maps or use specific scales tailored to individual maps to highlight finer variations, depending on the intended focus of the analysis. This flexibility allows for targeted exploration of irradiance patterns. The primary goal is to examine how light distribution occurs under typical conditions and how it evolves over different temporal scales. Both qualitative and quantitative evaluations are essential, as they provide insights into light availability and uniformity. These insights are crucial for informing APV system design, where light serves as the primary energy source and a dominant influencing factor. A detailed analysis of the reference configurations outlined in [Section 3.2](#), specifically Tracker 2P (interspace) and Fixed 2P (elevated), is conducted. Additionally, sensitivity simulations encompassing all specified configurations are presented to further refine the understanding of system performance under varying conditions.

Step 1 - Hourly Analysis: The initial phase of the irradiance analysis centers on hourly data to assess intra-day variability. Solar irradiance curves and heat maps are presented for selected hours (9:00, 11:00, 13:00, and 16:00) of the June Typical Meteorological Day (TMD, 30-06). These charts illustrate the irradiance incident on the

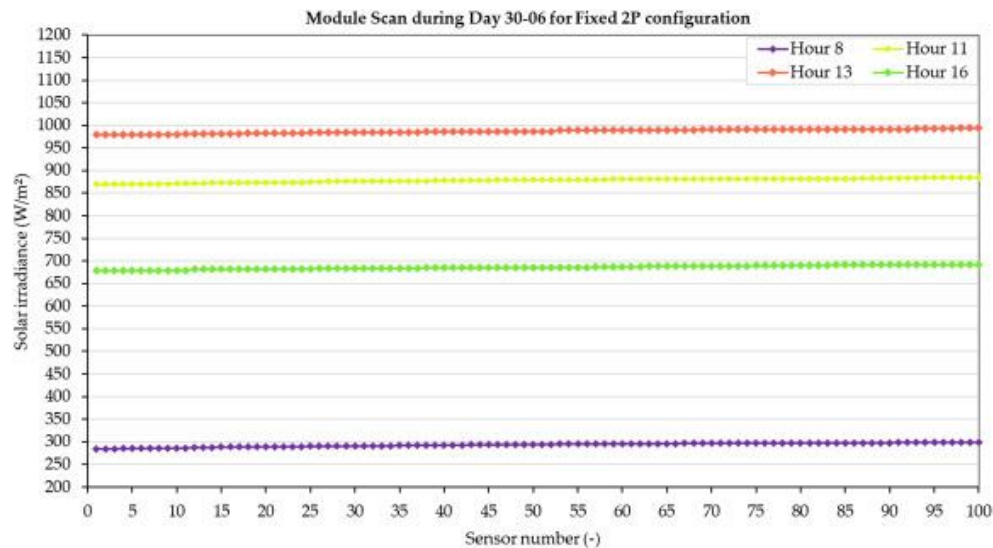
PV modules (*modulescan* function, Fig. 5, Fig. 6), followed by the irradiance at ground level (*groundscan* function, Fig. 7, Fig. 8). For module scans, sensors are positioned on the surfaces of the modules and are arranged vertically (along the zenith direction) at each simulation step.



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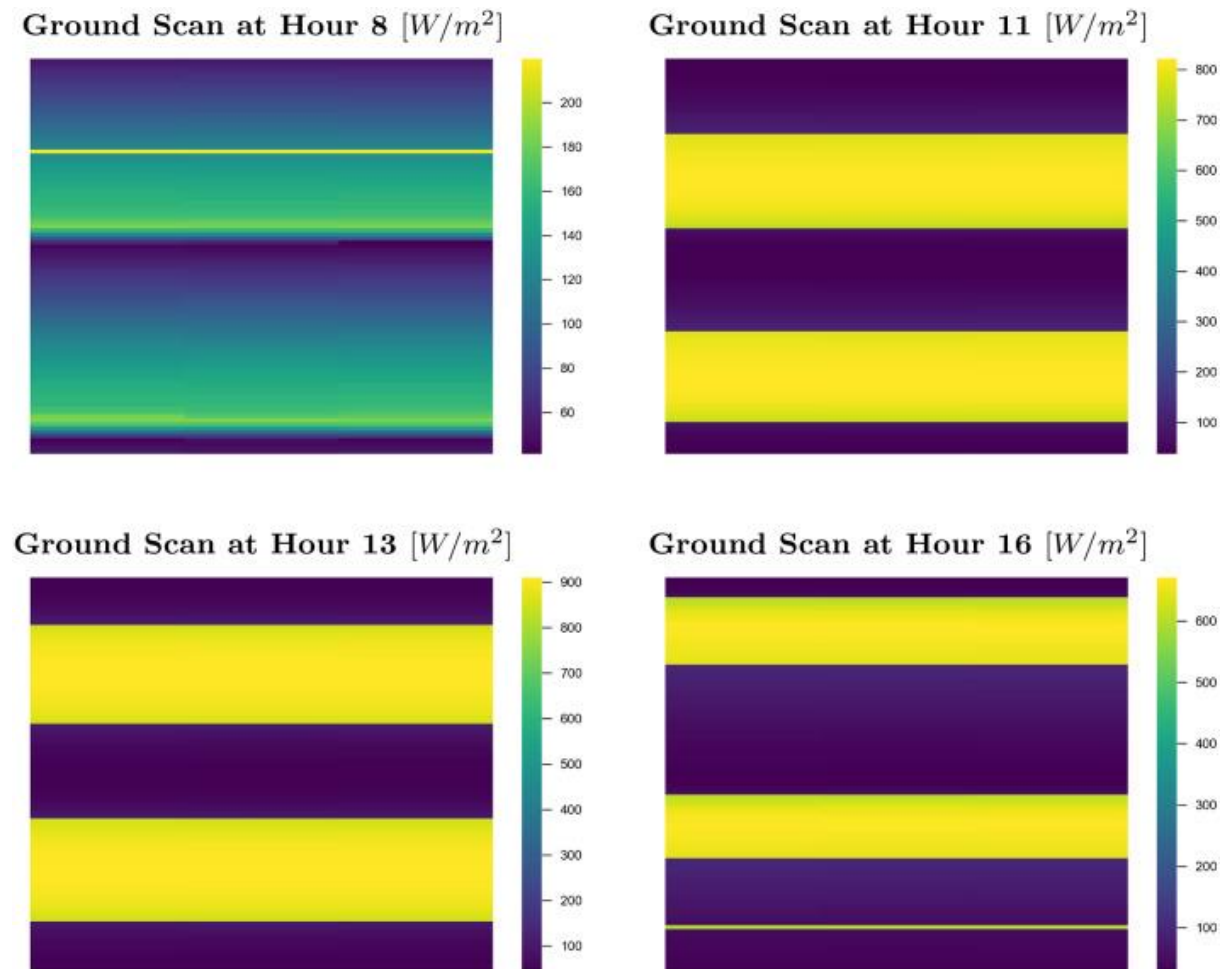
Fig. 5. Solar irradiance reaching the sensor covering one PV module at Hours 8, 11, 13 and 16 during 30-06 TMD for Tracker 2P configuration.



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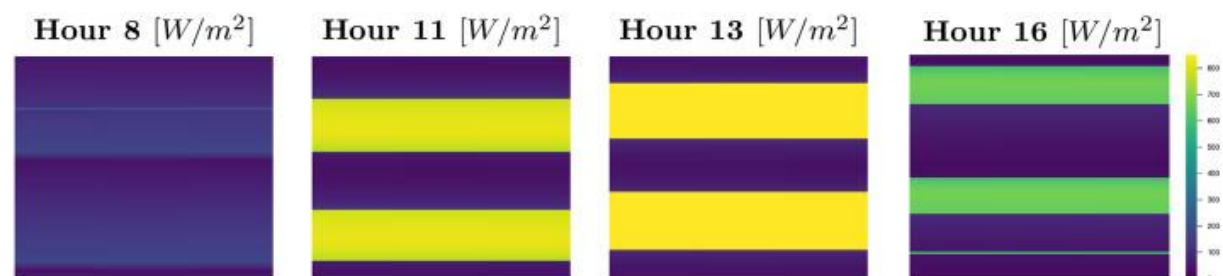
Fig. 6. Solar irradiance reaching the sensor covering one PV module at Hours 8, 11, 13 and 16 during 30-06 TMD for Fixed 2P configuration.



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Fig. 7. Solar irradiance reaching the ground on 30-06 TMD for Tracker 2P configuration. East is at the top.



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Fig. 8. Solar irradiance reaching the ground on 30-06 TMD for Tracker 2P configuration with a fixed measurement scale. East is at the top.

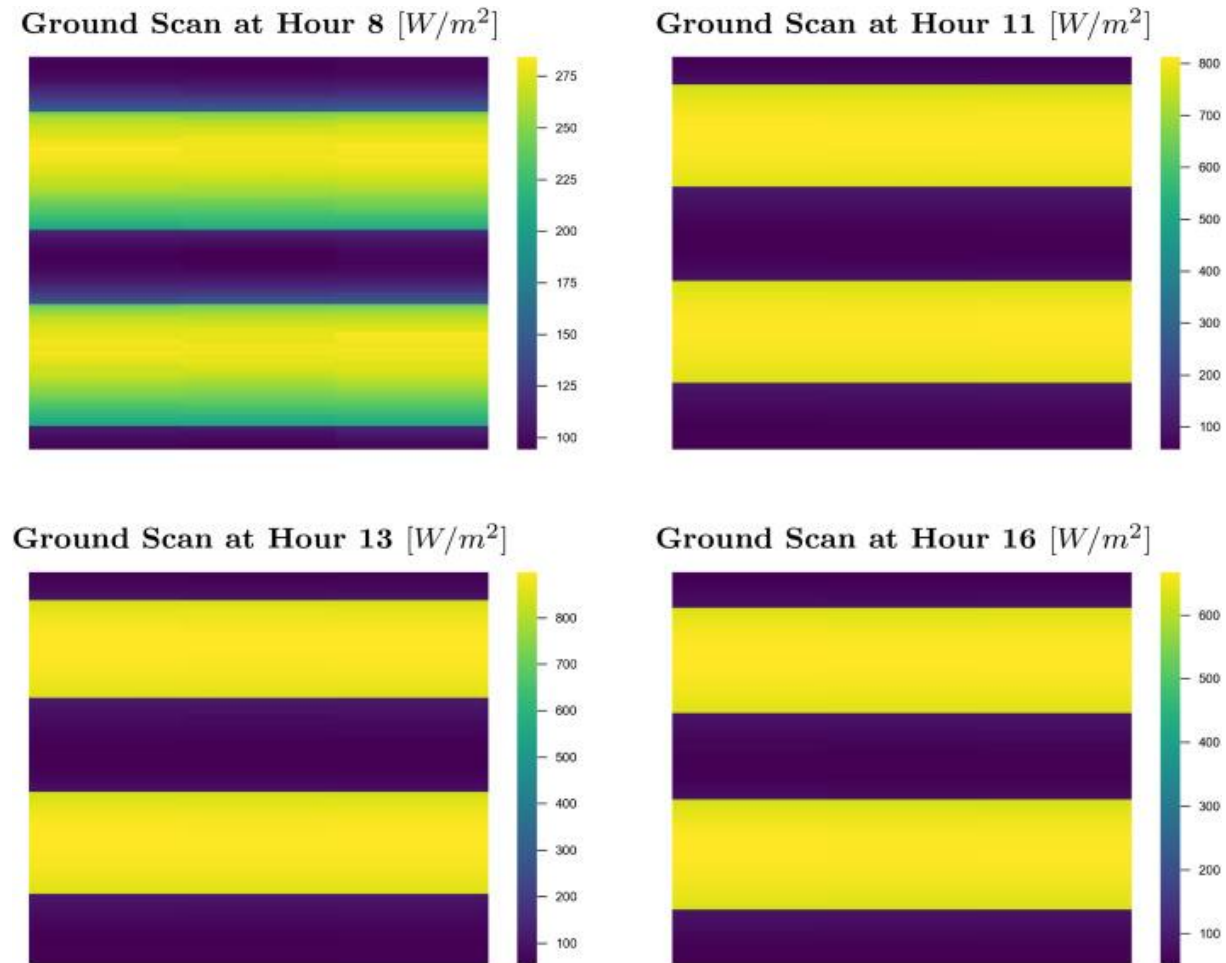
In the Tracker configuration, solar irradiance values that are registered by the sensors positioned in the space gap between the upper and lower modules in the 2-portrait configuration have been excluded and are not shown in the figure. No other significant pattern is observed and the radiation distribution across the module surfaces is largely uniform.

In the Fixed configuration, although the radiation distribution across the module surfaces is largely uniform, it exhibits more pronounced variations throughout the hours of the day when compared to the Tracker configuration.

For ground scans, irradiance distribution heat maps for the same date and hours are presented for the Tracker 2P and Fixed 2P reference configurations. Similar to the module scans, these heat maps initially use a scale specific to each map, enabling a detailed comparison of irradiance differences between adjacent sensors. This approach is particularly informative during early and late hours, when direct irradiance is weaker, and diffuse radiation contributes more significantly. Subsequently, a side-by-side comparison is provided using a uniform scale across all heat maps. While this approach flattens the gradients, it clearly highlights the overall trends of increasing and decreasing irradiance from the start to the end of the day. The heat maps reveal distinct dark areas along the width of the collector, corresponding to the shadows cast by module rows. No significant gradients are observed along the row direction, as the sensor-covered area is centrally located within the total APV layout, ensuring uniform module and row distribution in the vicinity. The shading is notably influenced by both the central gap and the torque tube. The Tracker configuration is characterized by horizontal stripes in the heat maps, a feature less prominent in the Fixed configuration. These horizontal variations, often appearing as noise, are indicative of the dynamic shading patterns associated with the Tracker system. This distinction between configurations is evident not only in the heat maps but also in other aggregated results.

[Fig. 9](#), [Fig. 10](#) illustrate the irradiance distribution for the Fixed 2P configuration. The modules are oriented with an azimuth angle of 0 (south-facing), consistent with standard PV systems in the Northern Hemisphere. This orientation contributes to significant heterogeneity in ground-level irradiance, which can result in localized microclimatic variations, such as differences in humidity and temperature. To mitigate this heterogeneity, adjusting the system orientation slightly towards the east or west could

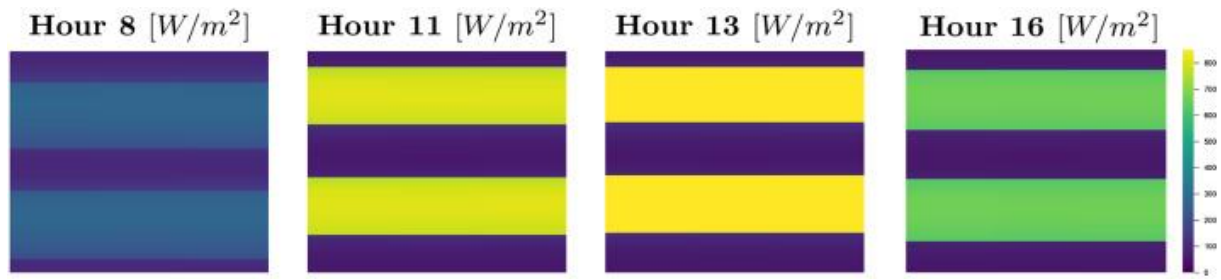
be explored, particularly during periods with high solar elevation, such as spring and summer. However, this suboptimal adjustment would lead to a reduction in the overall electrical energy yield of the PV system due to deviations from the optimal azimuth angle.



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Fig. 9. Solar irradiance reaching the ground on 30-06 TMD for Fixed 2P configuration. North is at the top.

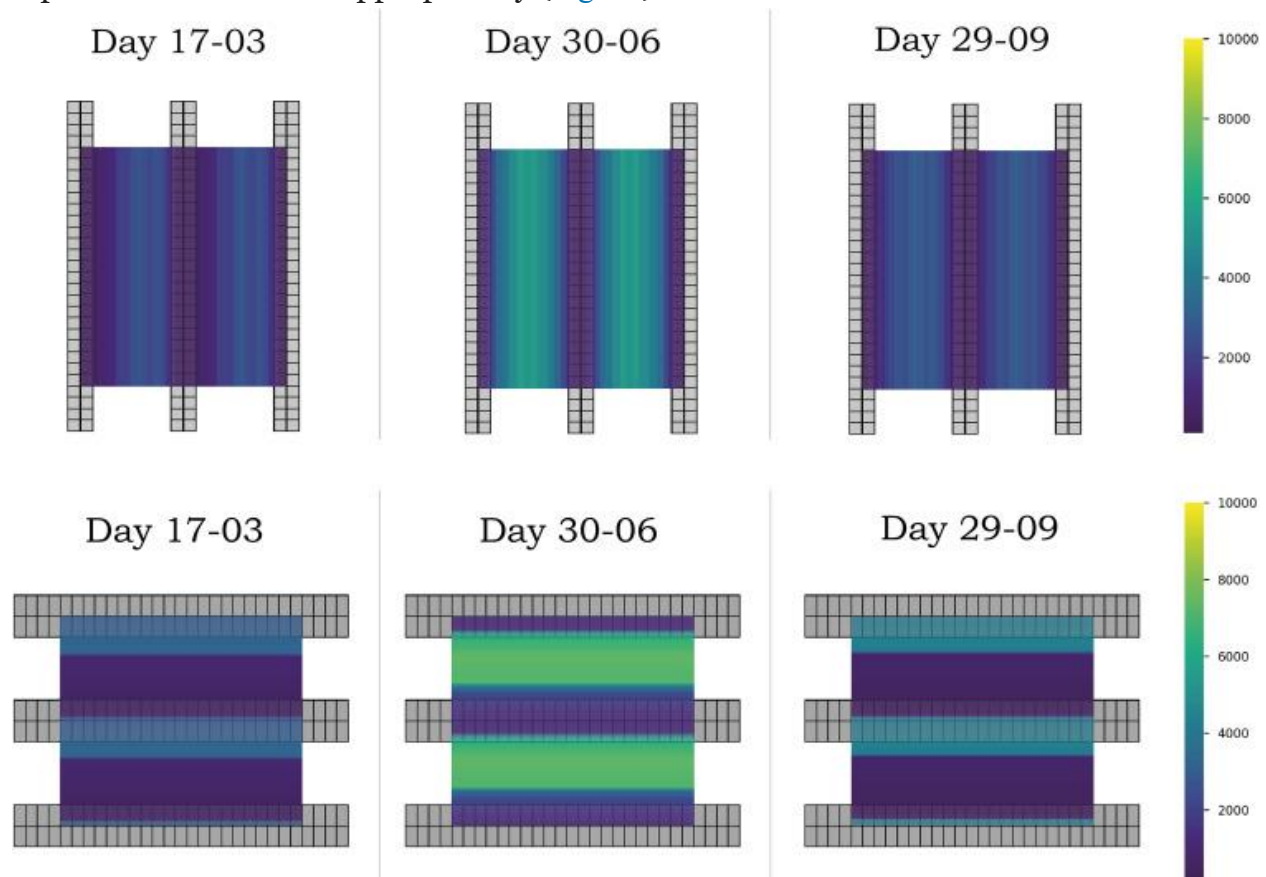


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Fig. 10. Solar irradiance reaching the ground on 30-06 TMD for Fixed 2P configuration with a fixed measurement scale. North is at the top.

Step 2 - Daily Analysis: The second step of the analysis examines daily data to assess inter-day variability. Simulation results for the March TMD (17-03), June TMD (30-06), and September TMD (29-09) are aggregated and presented as daily insolation values (in Wh/m). The corresponding heat maps are stacked on sections of the simulated scenes' top views and oriented appropriately (Fig. 11).



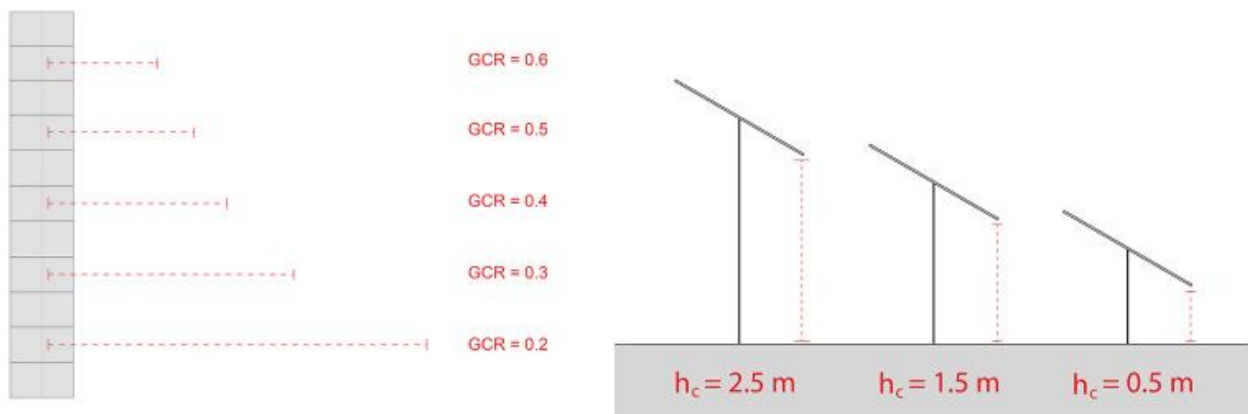
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Fig. 11. Solar insolation reaching the ground on 17-03, 30-06 and 29-09 for Tracker 2P (top) and Fixed 2P (bottom) configurations with a fixed measurement scale. North is at the top.

A notable dip in insolation along the trajectory of the module rows is observed for all TMDs. The sun's geometry influences the shading patterns on the ground differently across seasons, with shorter shadows in the summer due to the sun's higher elevation. This effect is particularly pronounced in the Fixed 2P configuration, where the variation in sun elevation causes noticeable shifts and translations in the shaded ground areas. Although the Tracker solution partially mitigates this effect, the sun's azimuth and elevation still significantly impact the shading throughout different seasonal simulations. In addition to the changes in shadow shape and depth, the reduction in sunlight hours and average GHI leads to a general decrease in insolation for March and September. This reduction should be carefully considered in terms of photosynthetically active radiation (PAR) and light saturation point (LSP) to better understand crop responses. The PAR is defined as the part of electromagnetic radiation that can be used as the energy source for plants' photosynthesis process, while the LSP represents a critical physiological threshold in photosynthesis, beyond which an increase in light intensity does not lead to a corresponding increase in the rate of photosynthesis. LSP generally changes not only within the species but also for different parts of the same crop, for different life stages, and for different climate conditions.

Step 3: Yearly and Sensitivity Analysis: A sensitivity analysis was conducted to examine yearly average insolation across 60 configurations, varying (0.2, 0.3, 0.4, 0.5, 0.6) and clearance height (0.5, 1.5, 2.5 m) for 2P and 1P cases. Fig. 12 illustrates these variations.

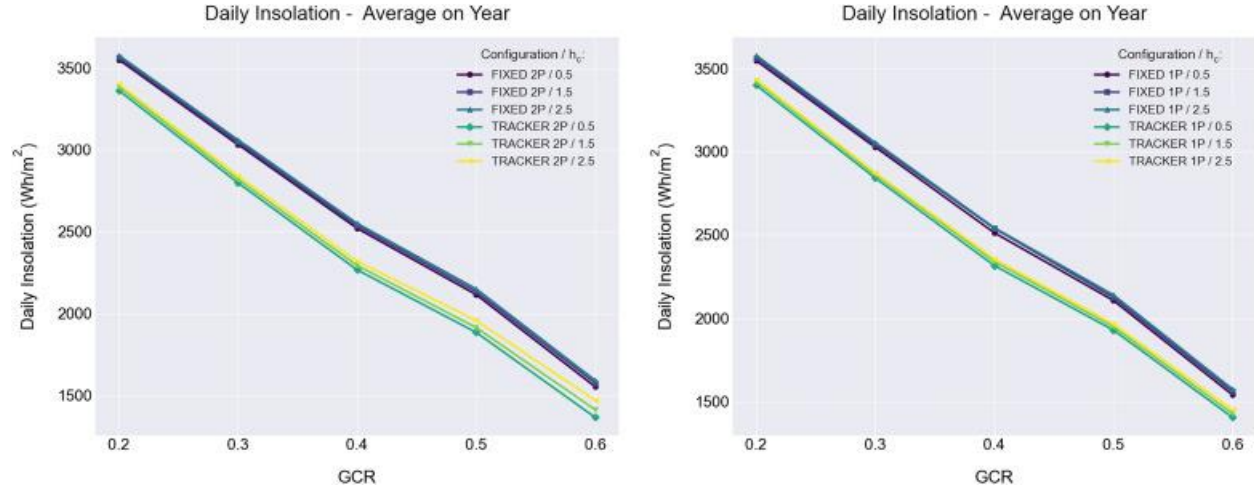


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Fig. 12. Variations of the parameters for sensitivity analysis.

Yearly average daily insolation is plotted for all configurations in Fig. 13. While clearance height has a limited effect on total insolation, it improves homogeneity at greater heights. Conversely, strongly impacts insolation, with higher causing steeper reductions due to mutual shading, consistent with previous studies [28].



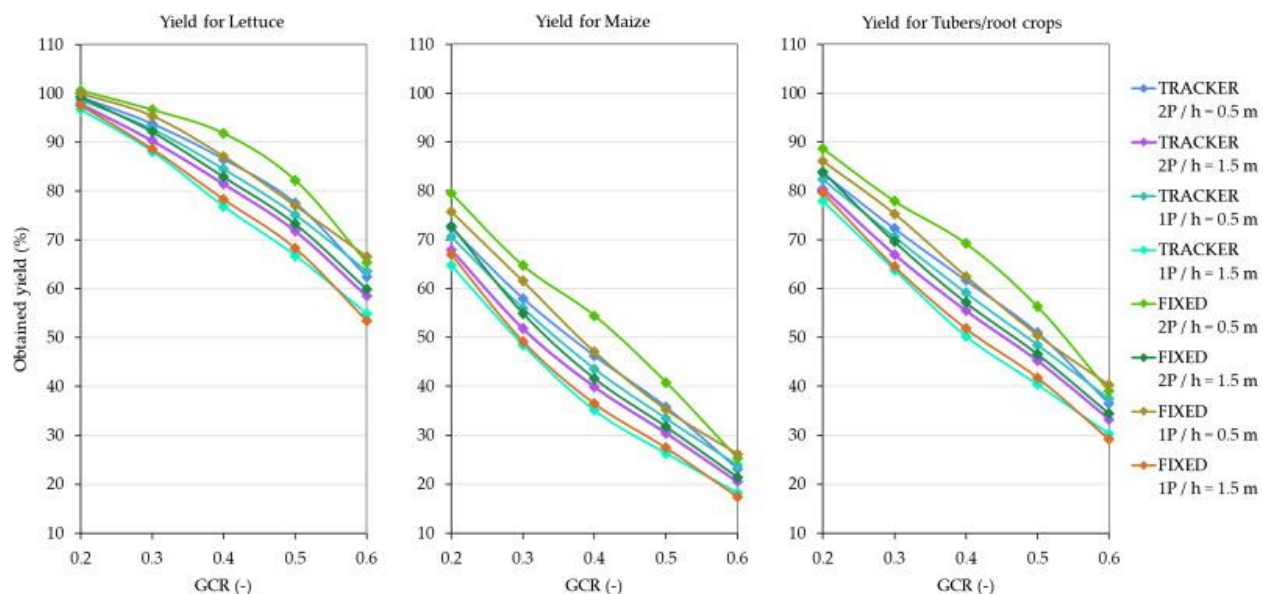
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Fig. 13. Daily insolation plot for all 2P (left) and 1P (right) configurations.

4.2. Crop yield assessment

The mean Reduction in Solar Radiation (RSR), derived from the ground scan and aggregated according to PV system and crop system definitions, is used to estimate RPY by means of the spline interpolation curves for crops, as previously illustrated in Fig. 4. The resulting data are shown in Fig. 14 for all interspace configurations.



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Fig. 14. Relative plot yield (RPY) for all crops and interspace configurations.

Once the lost land ratio is factored in, also the RFY is obtained for all configurations (see [Table 1](#)).

Table 1. Relative field yield (RFY) for all crops and configurations.

(m)	Configuration	GCR (-)				
		0.2	0.3	0.4	0.5	0.6
(-)						
0.5	Tracker 2P	0.82	0.69	0.56	0.45	0.30
1.5	Tracker 2P	0.81	0.67	0.53	0.42	0.28
2.5	Tracker 2P	0.92	0.83	0.72	0.64	0.53
0.5	Tracker 1P	0.82	0.69	0.55	0.44	0.30
1.5	Tracker 1P	0.80	0.65	0.50	0.39	0.26
2.5	Tracker 1P	0.92	0.83	0.73	0.64	0.52
0.5	Fixed 2P	0.84	0.72	0.60	0.48	0.31
1.5	Fixed 2P	0.82	0.68	0.54	0.43	0.29
2.5	Fixed 2P	0.98	0.85	0.74	0.64	0.51
0.5	Fixed 1P	0.83	0.71	0.57	0.45	0.32

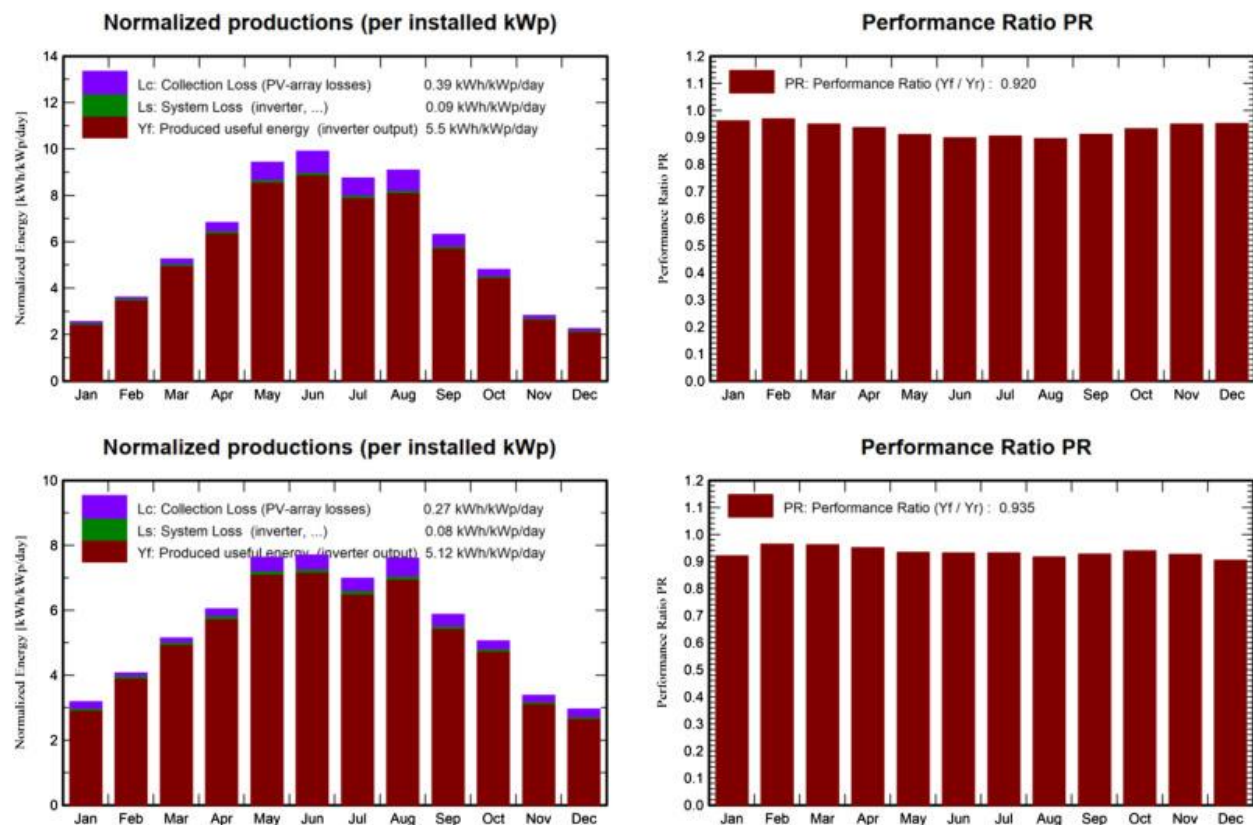
1.5	Fixed 1P	0.81	0.66	0.51	0.40	0.26
2.5	Fixed 1P	0.93	0.85	0.73	0.64	0.50
(-)						
0.5	Tracker 2P	0.60	0.43	0.30	0.21	0.11
1.5	Tracker 2P	0.56	0.38	0.26	0.18	0.10
2.5	Tracker 2P	0.58	0.43	0.32	0.25	0.18
0.5	Tracker 1P	0.59	0.41	0.28	0.19	0.11
1.5	Tracker 1P	0.54	0.36	0.23	0.15	0.09
2.5	Tracker 1P	0.59	0.44	0.32	0.25	0.17
0.5	Fixed 2P	0.66	0.48	0.35	0.24	0.12
1.5	Fixed 2P	0.60	0.41	0.27	0.18	0.10
2.5	Fixed 2P	0.77	0.46	0.33	0.25	0.16
0.5	Fixed 1P	0.63	0.46	0.31	0.20	0.13
1.5	Fixed 1P	0.56	0.36	0.24	0.16	0.08
2.5	Fixed 1P	0.61	0.46	0.33	0.25	0.16
(-)						
0.5	Tracker 2P	0.69	0.54	0.40	0.30	0.18
1.5	Tracker 2P	0.67	0.50	0.36	0.26	0.16
2.5	Tracker 2P	0.72	0.58	0.46	0.39	0.29
0.5	Tracker 1P	0.68	0.52	0.38	0.28	0.18
1.5	Tracker 1P	0.65	0.47	0.33	0.23	0.15
2.5	Tracker 1P	0.73	0.59	0.47	0.39	0.29
0.5	Fixed 2P	0.74	0.58	0.45	0.33	0.19
1.5	Fixed 2P	0.70	0.52	0.37	0.27	0.17
2.5	Fixed 2P	0.86	0.61	0.48	0.38	0.28
0.5	Fixed 1P	0.71	0.56	0.41	0.29	0.19
1.5	Fixed 1P	0.66	0.48	0.34	0.24	0.14
2.5	Fixed 1P	0.74	0.61	0.48	0.38	0.27

The obtained yield values show consistent variations across both height and GCR for all cases. Overhead systems are clearly better suited and favor crop production for multiple reasons. Interspace systems suffer from lost area due to PV footprint, with higher GCR severely impacting the yield. On average, the three crops show different behavior. In this sense, lettuce can be considered the most shade-tolerant crop, while maize is the most shade-susceptible crop.

Applicable insight can be gained if a minimum crop yield is set. If, for instance, a minimum of 50 % is required, even for the most shade-tolerant crop, a number of cases must be rejected. By looking at lettuce, this would mean excluding all interspace configurations for GCR of 0.5 and 0.6. Similarly, if a minimum of 70 % is required, the only viable option is a well-spaced overhead design.

4.3. PV yield assessment

All the configurations are simulated in PVsyst with the parameters previously described. The results in terms of normalized production and performance ratio are shown in Fig. 15 for Tracker 2P and Fixed 2P reference cases.



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Fig. 15. Normalized productions and performance ratio for reference Tracker and Fixed 2P configurations.

The specific production (kWh/kW), which is the parameter of main interest, and the PV yield are obtained for all reported in Table 2. The PV yield used for the final APV performance assessment is found as the ratio between the specific production for the selected APV configuration and the standard PV system that would be built on the same site based on current technologies and practices, see Table 3. Since the case studies of this research work also explore APV design solutions that follow today's PV common practices, values of equal or close to 1 are found. This means that the APV design in question is only slightly affected by crop presence, which is usually the case for interspace systems with GCR close to 0.5.

Table 2. Specific production.

(m)	Specific production (kWh/kW/y)				
	GCR (-)	GCR (-)	GCR (-)	GCR (-)	GCR (-)
	0.2	0.3	0.4	0.5	0.6
TRACKER 2P					
0.5	2236.46	2166.00	2081.11	2008.52	1894.15
1.5	2257.19	2182.85	2096.04	2021.00	1900.67
2.5	2270.19	2194.63	2105.85	2027.30	1902.19
TRACKER 1P					
0.5	2244.96	2176.04	2094.96	2025.70	1916.96
1.5	2271.18	2200.26	2115.22	2039.48	1921.81
2.5	2283.85	2210.78	2119.11	2040.11	1921.07
FIXED 2P					
0.5	1871.24	1861.81	1849.41	1834.41	1779.70
1.5	1903.34	1891.93	1875.93	1856.26	1795.15
2.5	1924.20	1911.50	1891.19	1868.11	1801.81
FIXED 1P					
0.5	1886.60	1874.44	1859.33	1836.11	1742.56

1.5	1936.85	1914.42	1891.04	1861.15	1757.37
2.5	1948.70	1926.09	1898.34	1865.24	1758.51

Table 3. PV relative yield.

(m)	PV relative yield (-)				
	GCR (-)	GCR (-)	GCR (-)	GCR (-)	GCR (-)
	0.2	0.3	0.4	0.5	0.6
TRACKER 2P					
0.5	1.11	1.08	1.04	1.00	0.94
1.5	1.12	1.09	1.04	1.01	0.95
2.5	1.13	1.09	1.05	1.01	0.95
TRACKER 1P					
0.5	1.11	1.07	1.03	1.00	0.95
1.5	1.12	1.09	1.04	1.01	0.95
2.5	1.13	1.09	1.05	1.01	0.95
FIXED 2P					
0.5	1.02	1.01	1.01	1.00	0.97
1.5	1.04	1.03	1.02	1.01	0.98
2.5	1.05	1.04	1.03	1.02	0.98
FIXED 1P					
0.5	1.03	1.02	1.01	1.00	0.95
1.5	1.05	1.04	1.03	1.01	0.96
2.5	1.06	1.05	1.03	1.02	0.96

It should be noted that the balance of plant (BOP) equipment and relative losses are considered secondary and are not factored in PVsyst simulations, since the needed PV yield is a relative yield and is not used to assess the production in general terms.

4.4. Performance metrics

LER values obtained for all PV configurations and for all crops are reported below ([Table 4](#)). An LER of more than 1 indicates that producing energy and crops together is more productive than producing them separately. Conversely, when LER is less than 1, it means that the APV system is less productive than separate production.

Table 4. Land Equivalent Ratio (LER) for Lettuce, Turnip, and Maize under various configurations.

(m)	Configuration	GCR (-)				
		0.2	0.3	0.4	0.5	0.6
0.5	Tracker 2P	1.94	1.77	1.60	1.45	1.24
1.5	Tracker 2P	1.94	1.76	1.57	1.42	1.23
2.5	Tracker 2P	2.05	1.92	1.77	1.65	1.48
0.5	Tracker 1P	1.93	1.76	1.58	1.44	1.25
1.5	Tracker 1P	1.92	1.74	1.54	1.39	1.21
2.5	Tracker 1P	2.05	1.93	1.77	1.65	1.47
0.5	Fixed 2P	1.86	1.73	1.61	1.48	1.28
1.5	Fixed 2P	1.86	1.71	1.56	1.44	1.27
2.5	Fixed 2P	2.03	1.89	1.77	1.66	1.49
0.5	Fixed 1P	1.86	1.73	1.58	1.45	1.27
1.5	Fixed 1P	1.86	1.70	1.54	1.41	1.21
2.5	Fixed 1P	1.99	1.89	1.77	1.65	1.46
0.5	Tracker 2P	1.71	1.51	1.34	1.21	1.05
1.5	Tracker 2P	1.69	1.47	1.30	1.18	1.04
2.5	Tracker 2P	1.72	1.53	1.37	1.26	1.12
0.5	Tracker 1P	1.69	1.49	1.32	1.19	1.06
1.5	Tracker 1P	1.66	1.45	1.27	1.16	1.04

2.5	Tracker 1P	1.72	1.53	1.37	1.26	1.12
0.5	Fixed 2P	1.68	1.49	1.36	1.24	1.09
1.5	Fixed 2P	1.64	1.44	1.29	1.20	1.08
2.5	Fixed 2P	1.82	1.50	1.36	1.27	1.15
0.5	Fixed 1P	1.66	1.48	1.32	1.20	1.07
1.5	Fixed 1P	1.61	1.41	1.27	1.17	1.04
2.5	Fixed 1P	1.67	1.51	1.36	1.26	1.12

0.5	Tracker 2P	1.81	1.61	1.44	1.30	1.12
1.5	Tracker 2P	1.79	1.58	1.40	1.27	1.11
2.5	Tracker 2P	1.85	1.68	1.51	1.39	1.24
0.5	Tracker 1P	1.79	1.60	1.42	1.28	1.13
1.5	Tracker 1P	1.77	1.56	1.37	1.24	1.09
2.5	Tracker 1P	1.85	1.68	1.52	1.39	1.24
0.5	Fixed 2P	1.76	1.59	1.46	1.33	1.16
1.5	Fixed 2P	1.73	1.55	1.39	1.28	1.14
2.5	Fixed 2P	1.91	1.65	1.51	1.40	1.26
0.5	Fixed 1P	1.74	1.58	1.42	1.29	1.14
1.5	Fixed 1P	1.72	1.52	1.37	1.26	1.10
2.5	Fixed 1P	1.80	1.66	1.51	1.40	1.23

The APV system's enhanced ability to capture radiation is evident. The lowest reported value is 1.04, while 2.05 is the highest. Other data from various European countries report values broadly in line with this study, ranging from 1.3 to LER exceeding 1.5 [8]. These are however highly dependent on the meteorological data considered (typical or time series), location and especially configurations. For the TMY and location under investigation, the PV production is only slightly affected by APV design and therefore provides the majority share of LER. Without introducing boundaries to productivity share, crops will be heavily affected by RSR especially lost land. Crop yields never reach unity. Moreover, it is observed that a lower GCR and hence a higher pitch return higher productivity across all cases. This is because both PV and crop systems benefit from mutual shading and crop shading decreases.

5. Conclusions

This study aimed to address several critical gaps in the design, modeling, and implementation of open-field APV systems, particularly focusing on optimizing them for both solar energy generation and agricultural productivity. By developing a comprehensive simulation framework, the study evaluated the impact of key design parameters on the performance of APV systems in temperate climates. The objectives were to develop a detailed simulation model for solar irradiance distribution, assess crop yield and energy production, conduct sensitivity analyses of design factors, and provide actionable insights for the optimization of APV systems in real-world scenarios, particularly in Europe and Italy.

The qualitative results demonstrated that design parameters, such as clearance height and GCR, have a significant influence on both crop yield and solar energy generation. Increasing the clearance height and adjusting the GCR were found to significantly reduce shading effects on crops, thus enhancing agricultural productivity, especially for shade-sensitive crops like maize. In contrast, shade-tolerant crops such as lettuce benefitted from the partial shading provided by APV systems, which also improved water use efficiency, protected crops from extreme weather events, and optimized the microclimate. The results confirmed the potential of APV systems to provide synergistic benefits, where land can simultaneously produce both food and renewable energy. The study also showed that bifacial PV modules, when combined with tracking systems, optimized solar energy generation by allowing the panels to adjust their orientation throughout the day to capture more sunlight, thus improving the overall energy output of the system.

Quantitative results revealed that system height and GCR are critical design factors that influence both energy and agricultural outcomes. For instance, when the GCR was set to 0.3 and the system height was increased to 2.5 meters, crop yields for maize were approximately 25 % higher compared to configurations with lower clearance and higher GCRs. Furthermore, energy yield from APV systems with single-axis tracking technology increased by 18 % compared to fixed systems, confirming the advantages of tracking systems in optimizing energy production. The LER values, ranging from 1.04 to 2.05 across various APV configurations, indicated that the integrated APV systems consistently outperformed conventional separate land-use models, demonstrating that APV systems can offer enhanced land-use efficiency. The sensitivity analysis highlighted that system height had the most significant impact on overall performance, followed by GCR while tracking technology further improved solar energy generation, though at the cost of slightly reduced crop yields due to increased shading during certain periods of the day.

Looking ahead, the study highlights several areas for future research and development. Refinement of simulation models to include a wider range of crops, particularly those with permanent or perennial growth patterns, is crucial, as these crops are less studied in the context of APV systems. In addition, field validation of the simulation results is essential to confirm the applicability of the proposed designs under real-world conditions. Future research should also focus on the economic feasibility of APV systems, assessing the impact of policy environments and financial incentives on the scalability and implementation of APV projects.

An additional factor that can influence the long-term performance of APV systems is the impact of environmental conditions such as dust accumulation, wind storms, and heavy rainfall. These interactions highlight the need for further research into integrated design approaches that optimize system resilience while maintaining energy and agricultural productivity. Future studies should explore predictive maintenance strategies and adaptive control systems that respond dynamically to environmental conditions, ensuring the reliability and sustainability of APV installations. In conclusion, this research provides valuable information on the design and optimization of APV systems, confirming their potential to contribute significantly to sustainable land use, solar energy generation, and agricultural productivity.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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CRedit authorship contribution statement

Domenico Mazzeo: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Formal analysis, Data curation, Conceptualization. **Andrea Di Zio:** Writing – original draft, Software, Investigation,

Data curation, Conceptualization. **Claudio Pesenti:** Supervision, Resources, Methodology, Formal analysis, Conceptualization. **Sonia Leva:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Formal analysis, Conceptualization.